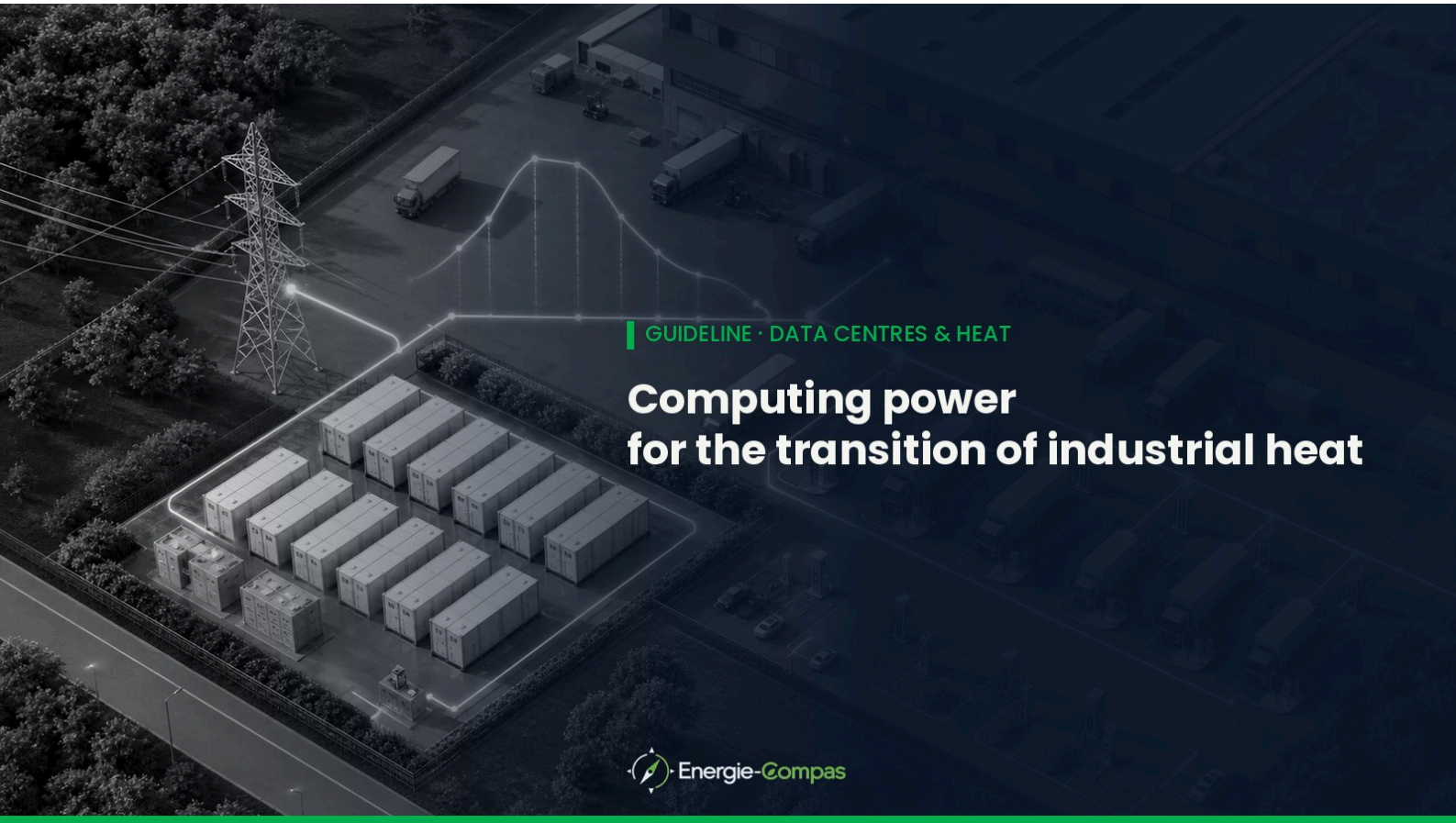
Energie-Compas



GUIDELINE · OCTOBER 2026

Inference & Industrial Heat

Data centres up to 10 MW as a driver for the cost-effective electrification of industrial heat. A guideline for conversations with investors and developers, and a concise manual for industrial companies.

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1. Why this guideline

An inference data centre of 5–10 MW is essentially a 5–10 MW electric heat source that delivers computing along the way.

This guideline turns that property into a design principle: data centres are built where their heat replaces fossil industrial heat, and their grid connection also serves the electrification of that industry.

Opposition to data centres is justified when they absorb grid capacity, water and renewable electricity without giving anything back. Spain and Germany are responding with stricter standards and thresholds. This guideline takes a third route: **integration instead of exclusion**. A data centre that supplies its heat to an industrial neighbour spreads the cost of a heavy grid connection over two useful purposes.

Why up to 10 MW

- Inference workloads run best close to the user and can be built modularly in blocks of 1–5 MW.
- 5–10 MW of residual heat matches the heat demand of a single industrial site or a small heat network.
- A connection of up to 10–25 MVA stays within the reach of distribution or local transmission grids, without new custom high-voltage substations.
- This segment falls just below the new 10 MW threshold in Spain, and is already covered by the law in Germany. So the point is not stricter rules, but a framework that is future-proof.

Who it is for

Investors and developers

Chapters 2, 3, 4 and 6 show how we select sites and which standards we test them against.

Industrial companies

Chapter 5 is a quick self-test. Anyone who wants to know whether their heat can be electrified cost-effectively with a data centre as the source can make a first estimate there in half an hour.

If this sounds like you, we invite you to get in touch (chapter 7).



2. Facts

Almost all the electricity a data centre consumes is released as heat. At 7 MW IT and 8,000 full-load hours, that is roughly 55 GWh of low-grade heat per year, available as base load.

Temperature levels

Cooling technology	Available temperature	Direct users	With heat pump
Air cooling	25–35 °C	aquaculture, greenhouse soil heating	up to 70–90 °C, COP ~3
Direct liquid cooling (DLC)	45–60 °C	greenhouses, low-temperature networks, preheating, cleaning	up to 90–120 °C, COP ~2.5–4
Immersion	45–55 °C	as DLC	as DLC

COP values are indicative and depend on the temperature lift. Modern AI hardware increasingly requires DLC, which makes the heat level attractive for industry.

Belgian context

- In 2025, 34% of Belgian generation came from nuclear, 34% from renewables and about 19% from gas; on top of that there were 14 TWh of net imports, mainly from France and the Netherlands (Elia via Brussels Times, Connaissance des Énergies).
- The carbon-free share of consumption is therefore roughly 50%. A data centre without contracts already scores about 50% CFE.
- Grid capacity is the scarcest factor. A heavy connection that only serves computing is hard to justify to society; one that also electrifies industrial heat is not.
- Flanders faces water scarcity: evaporative cooling on mains water is not an option.

Lessons from Spain

The Spanish draft royal decree, sent to the Council of State on 26 October 2026 (Reuters via The Star, El Periódico de la Energía):

- Threshold raised from 1 to 10 MW; reporting of water and electricity from 500 kW IT.
- 80% renewable per hour, with a transition to 2031 if the data centre accepts unpaid curtailment.
- PUE max. 1.25 per climate zone (was 1.15); WUE max. 0.1.[1]

> **Lesson:** hourly matching and grid flexibility as a condition are strong; unpaid curtailment and an 80% requirement from day one are not feasible for smaller players.

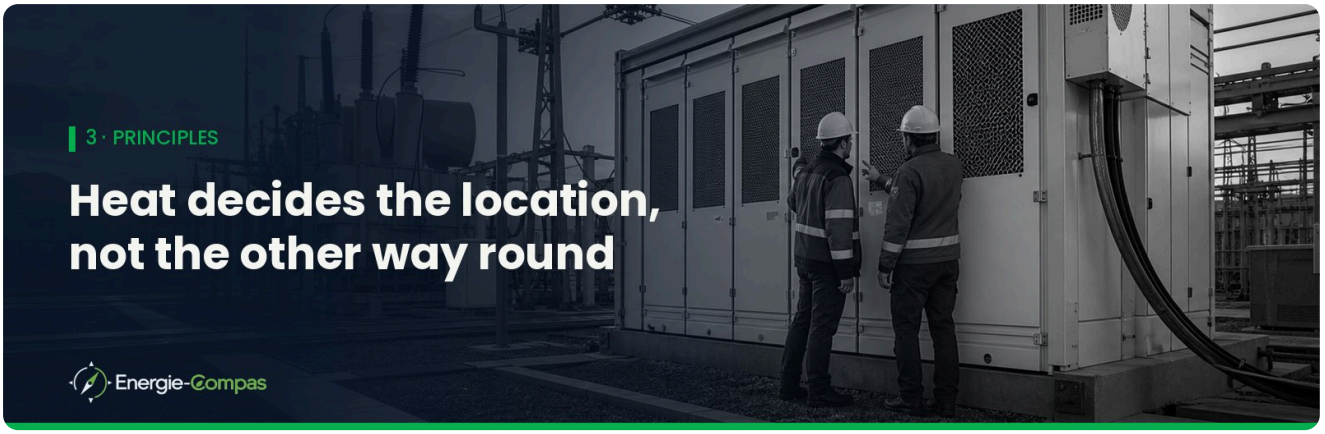
Lessons from Germany

The Energy Efficiency Act (EnEfG) applies from 300 kW (Gleiss Lutz, CMS):

- New data centres from 1 July 2026: PUE max. 1.2.
- Mandatory reuse of waste heat (ERF), rising to 20% depending on the commissioning date.

> **Lesson:** an ERF obligation does not work if the location has no heat user. An exemption for heat networks that are too small hollows out the obligation and pushes construction to greenfield sites (Öko-Institut). **Heat must be a siting criterion, not an after-the-fact obligation.**

[1] **PUE** (Power Usage Effectiveness) = total energy consumption of the data centre ÷ energy consumption of the IT equipment. A PUE of 1.25 means 25 kWh extra for cooling and losses per 100 kWh of IT. **WUE** (Water Usage Effectiveness) = annual water consumption in litres ÷ IT energy consumption in kWh. A WUE of 0.1 means 0.1 litre of water per kWh of IT.



3. Principles

This guideline rests on seven principles.

1. **One connection, two functions.** The grid connection is sized and contractually shared so that it also serves the electrification of the heat user (heat pumps, e-boilers).
2. **Heat strengthens the location.** A site with an identified industrial or community heat user gets priority. A letter of intent from that user before the grid connection request strengthens the case.
3. **Measured, not designed.** All performance is measured in operation, per quarter-hour where relevant, and not based on design values.
4. **Carbon-free per hour, grid-inclusive.** Carbon-free coverage is calculated per hour, counting the grid share and a rising share from own contracts.
5. **Grid-supportive behaviour.** The data centre accepts flexible connection terms in return for compensation and makes its battery and UPS available to the system.
6. **No mains water for cooling, no diesel as back-up.**
7. **Transparency.** Performance is reported publicly every year, and heat is offered to third parties nearby on transparent terms.



4. Target standards

We do not aim to be stricter than regulation, but future-proof: we take responsibility for the transition of the region where we compute.

That is why we apply these target standards to new sites of 1 to 10 MW IT. They are at least at the German level; the focus is on what the region gains: heat for local industry and hourly carbon-free coverage.

Parameter	Minimum at commissioning	Target 2027	Target 2030	Inspired by
PUE (annual average, measured)	≤ 1.30 in year 1	≤ 1.20	≤ 1.15	EnEfG 1.2; Spain 1.25
WUE mains water	≤ 0.1 l/kWh	≤ 0.1	≤ 0.05	Spain 0.1
ERF (share of energy reused externally)	≥ 10% + heat-ready	≥ 25%	≥ 50%	EnEfG up to 20%
CFE grid-inclusive (hourly, 3-year average)	≥ 50%	≥ 65%	≥ 80%	Spain 80% per hour
Flexible connection	accepted, compensated	same	same	Spain (unpaid)
Battery/UPS grid-supportive	≥ 30 min at IT load	active in grid support or congestion management	same	—
Back-up without fossil diesel	HVO or battery	same	same	—
Public reporting	annual	same	same	EU reporting, Spain ≥ 500 kW

Definitions

PUE = total data centre energy / IT equipment energy. Electricity for heat pumps that upgrade heat for external users is not counted.

ERF = energy usefully reused externally as heat / total data centre energy, measured with heat meters at the site boundary.

Heat-ready = high-temperature cooling (return ≥ 45 °C where the hardware allows), a heat exchanger with a connection point at the site boundary, and metering infrastructure, from day one.

CFE grid-inclusive, per hour h, with L = data centre consumption, C = contracted carbon-free generation in that hour (capped at L) and g = carbon-free share of the grid (consumption-based, including imports):

$$CFE_h = \frac{C_h + g_h \cdot (L_h - C_h)}{L_h} \qquad CFE_{year} = \frac{\Sigma_h \, CFE_h \cdot L_h}{\Sigma_h \, L_h}$$

With g ≈ 50%, reaching 65% requires about 30% hourly matching from own contracts, and 80% about 60%. Volumes the data centre claims through guarantees of origin or PPAs are not counted again in g.

Heat supply

- Heat is offered at a price that covers the additional cost of supply, not at gas-price parity plus a margin.
- Any third party within 3 km can request an offer; a refusal must be justified on technical or economic grounds.
- With a shared connection, the capacity split is fixed contractually in advance.



5. Feasibility for the industrial heat user

An industrial site can estimate in three steps whether a data centre is a feasible heat source: four knock-out criteria, a quick calculation and a score on nine parameters.

Step 1 – Knock-out criteria

If the site meets all four, the option is worth investigating.

- [ ] Continuous heat demand of at least 2 MW thermal for at least 6,000 hours a year, or seasonal storage (ATES/BTES) is possible.
- [ ] Most of the heat is needed below 120 °C.
- [ ] There is a building plot for the data centre within 3 km, preferably within 1 km.
- [ ] There is a realistic path to grid capacity: an existing connection that can be shared, or an indication from Fluvius or Elia within 3 years.

Step 2 – Rules of thumb

Recoverable heat, with P_{IT} = IT load, h = operating hours and f = recovery fraction (0.7–0.9 with DLC, ~0.5 with air cooling):

$$Q_{rec} = P_{IT} \times h \times f$$

After a heat pump with coefficient of performance COP:

$$Q_{useful} = Q_{rec} \times \frac{COP}{COP - 1} \qquad E_{HP} = \frac{Q_{rec}}{COP - 1}$$

Rule of thumb: 1 MW of IT with DLC delivers about 6–7 GWh of heat at 45–60 °C over 8,000 hours, and after upgrading to 90 °C (COP ~3.5) about 8–9 GWh of useful heat.

Cost comparison per MWh of heat (illustrative):

Item	Gas boiler	Data centre + heat pump
Energy	gas €35/MWh ÷ efficiency 0.9 = €39	electricity €100/MWh ÷ COP 3.5 = €29
CO_2	0.2 t/MWh × €75/t ÷ 0.9 = €17	—
Data centre residual heat	—	~70% of the heat × €5/MWh = €4
Capital cost heat pump + pipework	existing	€8–15
Total	**~€56/MWh**	**~€41–48/MWh**

Assumptions: gas price, ETS price, all-in electricity price and residual heat price are example values. Fill in your own tariffs; the electricity/gas price ratio and the COP are the most sensitive parameters.

Grid synergy: a site that electrifies its heat needs a heavier connection anyway. Shared with a data centre, the connection cost is split, but live inference is a fairly fixed load: users expect fast responses. Flexibility comes mainly from the battery and UPS. The data centre itself can contribute to a limited extent through GPU power capping, deferring batch inference (embeddings, evaluations) or shifting load to another site.

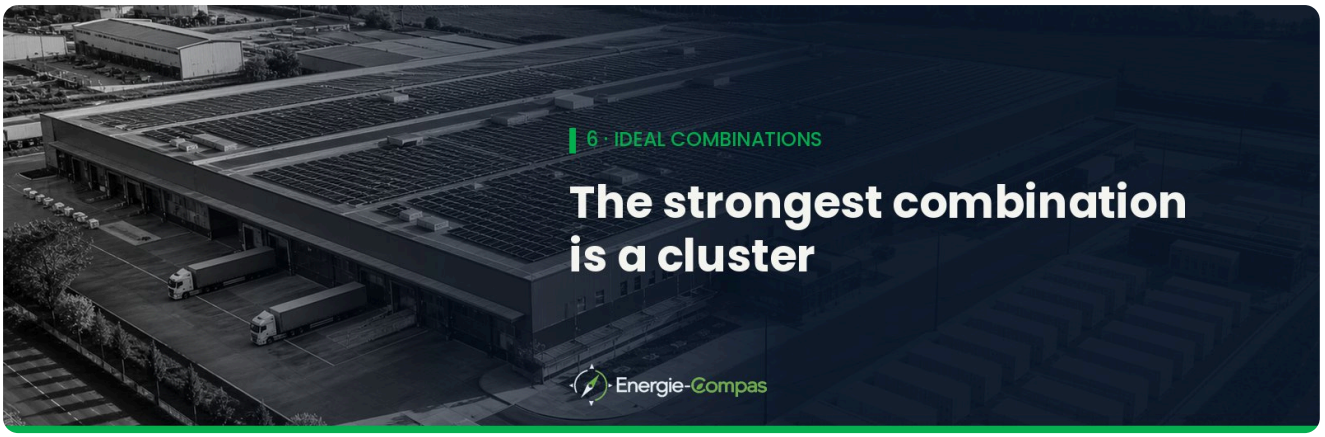
Land use: a liquid-cooled inference data centre takes up an indicative 250–500 m² of building per MW of IT (IT hall, electrical rooms, UPS/battery, cooling). With a medium-voltage substation, dry coolers and access, that comes to about 0.5–1 ha for a 10 MW site. Modular container solutions can be smaller; air-cooled designs need more.

Step 3 – Scoring model

Score each parameter from 0 to 3 and multiply by the weight. The maximum is 60 points.

Parameter	Weight	0	3
Base-load ratio: minimum hourly demand / data centre heat supply	3	< 25%	> 75%
Required temperature	3	> 120 °C	< 60 °C
Distance to data centre	2	> 3 km	< 500 m
Grid capacity available or shareable	3	> 5 years' wait	available
Current heat cost incl. CO_2	2	< €40/MWh	> €60/MWh
Replacement date of boiler or CHP	2	> 10 years	< 3 years
Space and permitting (industrial zone)	2	no space	industrial zoning, space available
Local CFE options (rooftop PV, wind, PPA pool)	1	none	wind + PV nearby
Seasonal storage ATES/BTES possible	2	no	yes, proven subsurface

Reading: ≥ 42 points = strong candidate, start a feasibility study · 30–41 = promising if one bottleneck is resolved · < 30 = pursue another decarbonisation route first.



6. Ideal combinations

The best sites combine continuous heat demand below 100 °C with a connection that needs upgrading anyway.

Archetype	Heat demand	Temp.	Base load	Why it works	Point of attention
Food industry (dairy, beer, vegetables)	preheating, cleaning, pasteurisation	60–90 °C	high, 5–7 days	large, continuous demand; boilers often due for replacement	shift patterns and weekend stops
Industrial laundry	hot water	60–80 °C	high	simple integration	smaller scale, 1–3 MW
Aquaculture	water heating	25–35 °C	very high	direct, no heat pump needed	niche market
Greenhouse horticulture	space heating, CO_2	40–60 °C	seasonal	large volume, CO_2 synergy when combined	summer surplus, storage needed
Chemicals and paper (preheating)	feed water for steam production	90–150 °C	very high	huge demand; part below 120 °C	high-temperature heat pump, lower COP
4th/5th-generation heat network + ATES	buildings, swimming pools, hospitals	40–70 °C	low, high with storage	public support, municipal heat plans	slower process, public partner
Vacated industrial site	variable	variable	n/a	existing heavy connection and zoning	heat user often still to be found

The strongest combination is a **cluster**: an industrial base-load user as anchor, complemented by a heat network or greenhouse for the rest, and seasonal storage to bridge the summer.



7. Let's explore together

This guideline is an invitation to talk, not an end point. If you have a site, a heat demand or capital that fits this story, we are happy to take the next step with you.

Industrial companies

- Score your site with chapter 5 and send us the result, or do it together with us.
- We review your heat demand, temperature levels and grid connection without obligation, and tell you honestly whether a data centre is the best option.

Investors and developers

- We share our selection criteria and discuss locations that meet the principles.
- We are glad to work together on the standards: which are feasible, which should be tighter.

Grid operators, municipalities and knowledge partners

Shared and flexible connections, heat plans and metering methods need coordination. We actively seek that dialogue.

What we are still working on

- Validating the grid-inclusive CFE method on 2025 quarter-hourly data.
- Cost comparison per archetype using current tariffs.
- A legal model for a shared connection between a data centre and industry.

Temperature levels, COP values, recovery fractions and the cost comparison are indicative industry values and example assumptions, not measured figures.

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Sources

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